

AGF

ANNUAL PERFORMANCE REPORT:

Agency: Arizona Game and Fish Department

Project: Section 6, E5-9

Project Title: Survey, monitor and reintroduce threatened and endangered species

Job Title: Razorback sucker and Colorado squawfish reintroduction and monitoring in the Verde and Salt rivers

Job Number: 36 (FCN5567)

Segment: 4 of 5 (July 1, 1997 to June 30, 1998)

Job Objective: evaluate survival of large sub-adult razorbacks and squawfish in the Salt and Verde rivers through stocking and monitoring of movements, and determine whether such stockings can assist in recovery efforts in these Lower Basin waters.

A. Summary of Progress:

Procedures

1. Propagation activities for the razorback sucker (*Xyrauchen texanus*) and Colorado squawfish (*Ptychocheilus lucius*) at Dexter National Fish Hatchery and Technology Center (DNFHTC) and Bubbling Ponds State Hatchery (BPSH) were directed at growing large individuals for stocking. A total of 1992 razorbacks and 2177 squawfish were stocked into the Verde River at Beasley Flat and Childs. All razorbacks and squawfish stocked during this segment were tagged with coded wire tags for stock identification. All squawfish received tags in the back area just under the dorsal fin on the right side of the fish. Razorbacks received tags in the back area just under the dorsal fin on the left side of the fish. Fifty-four (54) squawfish were PIT (Passive Integrated Transponder) tagged prior to release. Forty (40) razorbacks stocked into the Verde River were tagged with PIT tags. Many of both species still exhibited morphological anomalies. Though more common in the squawfish (50%) in the form of a "snake-head," several razorbacks (<10%) displayed "bent" caudal fins and deformed mouths. An infestation of Asian tapeworm also was detected in squawfish and treated at BPSH prior to release. Very low mortality (<2%) was experienced as a result of the infection and treatment. No tapeworms were detected in the razorbacks. Detailed information, such as individual weights and lengths, wire tag codes, and tagging location, are on file at Nongame Branch, Arizona Game and Fish Department (AGFD).

No fish were stocked into the Salt River in 1997-98, due to the White Mountain Apache Tribe's concerns about legal implications of stocking razorback suckers and Colorado squawfish. However, a meeting was arranged and attended by representatives of AGFD, the U.S. Fish and Wildlife Service, the San Carlos Apache Tribe and the White Mountain Apache Tribe on native fish management on the Upper Salt River. Future plans to resume stocking Colorado squawfish and other native species within Tribal lands are a possibility.

Table 1. Date, species, stocking location, numbers, size, and origin of fish stocked into the Verde River during FY97-98.

Date	¹ Species	Stocking Location	Number Stocked	Mean Length (TL)	Origin
970711	PTLU	Childs	33	477 mm	DNFHTC
970711	XYTE	Childs	765	287 mm	DNFHTC
971022	XYTE	Childs	320	392 mm	BPSH
971023	XYTE	Childs	556	394 mm	BPSH
971106	PTLU	Beasley Flats	500	445 mm	BPSH
971106	PTLU	Childs	1000	432 mm	BPSH
971110	PTLU	Beasley Flats	644	430 mm	BPSH
980223	XYTE	Childs	351	330 mm	BPSH

¹PTLU=*Ptychocheilus lucius*; XYTE=*Xyrauchen texanus*

2. Longitudinal monitoring surveys were conducted on the Salt River in August 1997 and June 1998. The Verde River was surveyed in July 1997, January and May of 1998. Collection gear used during these surveys included raft and canoe mounted electrofishers, seines, gill and trammel nets and angling.

SALT RIVER

On March 11, 12 and 13, 1998, we surveyed the Salt Arm of Roosevelt Lake using experimental gill nets. The lake was at 25 percent of maximum storage. Inflow from the Salt River was 4200 cfs. We used 12 45.7 m x 1.8 m gill nets, each with 6 panels of 2.5 cm to 7.6 cm mesh. A total of 44.5 net night units (NNU) were expended to catch 727 fish, for a catch per unit effort (CPUE) of 16.34 fish per net night unit (NNU = 1 standard net set from sunset to sunrise). Fish species and relative abundance were: common carp (*Cyprinus carpio* [29%]), largemouth bass (*Micropterus salmoides* [22.7%]), black crappie (*Pomoxis nigromaculatus* [21%]), channel catfish (*Ictalurus punctatus* [13.3%]), flathead catfish (*Pylodictus olivarius* [8.8%]), bluegill (*Lepomis macrochirus* [0.6%]), black buffalofish (*Ictiobus niger* [2.1%]), smallmouth bass (*Micropterus dolomieu* [1.4%]), redear sunfish (*Lepomis microlophus* [0.6%]), golden shiner (*Notemigonus crysoleucas* [0.3%]), and Sonora sucker (*Catostomus insignis* [0.3%]).

From August 20 through August 22, 1997, we surveyed the upper Salt River from Horseshoe Bend (RM 13.5) to Highway 288 bridge (river mile 0.6). River flow was 150 cfs and clear, conductivity exceeded 1990 μ S, and pH was 8.5. We caught 150 fish, all nonnatives (6 species). Flathead catfish represented 59.3 percent, followed by common carp (19.3%), channel catfish (16.7%), green sunfish (2%), smallmouth bass (1.3%), and red shiner (1.3%). Overall effort consisted of 202.5 minutes electrofishing, and 1 over-night trot line set (25 hooks). Total CPUE was 0.74 fish/minute and 4 fish/trot line set. No razorbacks or squawfish were collected. Ongoing tagging of flathead catfish was continued, to better understand the population dynamics of this species. Twenty-eight (28) flatheads were tagged and released. No recaptures occurred on this trip.

A second survey trip was conducted June 8-12, 1998, from Gleason Flat (RM 31.5) to Highway 288 Bridge (RM 0.6). Flows were 1000 cfs and turbid, conductivity was 1400 μ S, and pH was 8.6. The total catch was 523 fish; only 2 were native species. Overall effort consisted of 444 minutes electrofishing, and 2 NNUs of experimental gill net effort. Fish were collected by seine net, however effort could not be quantified due to loss of data in a rafting accident. CPUE was 1.2 fish/minute, and 2 fish/NNU. Species composition was led by flathead catfish (68.5%), followed by channel catfish (19.3%), common carp (11.3%), red shiner (*Cyprinella lutrensis* [0.6%]), and desert sucker (0.4%). Hundreds of red shiners

were taken, but seining data were lost at Quartzite Falls. We also took 2 fathead minnows (*Pimephales promelas*), 10 green sunfish (*Lepomis cyanellus*), and 1 common carp while seining. The survey objective was to recover previously stocked razorbacks and squawfish, as well as to tag flathead catfish to better understand population dynamics and movements in the upper Salt River system. We tagged and released 110 flathead; 8 had existing Floy tags.

We conducted a third trip from June 24 through June 26, 1998, from Horseshoe Bend to the Highway 288 bridge. Water clarity was good at 1.5 m, with a pH of 8.4, conductivity of 2430 μ S, and water temperature of 25°C. A total of 246 minutes electrofishing were expended to catch 387 fish. This equates to 1.6 fish/minute. A total of 265 flathead catfish were sampled, representing 68.5 percent of the catch. Channel catfish and common carp made up the rest of the sample. Thirty-one (31) flathead catfish were tagged with Floy tags. Eighteen (18) recaptured flatheads, (tagged in 1996, 1997, and 1998) were weighed, measured, and released. Almost no backwater habitats existed at flows experienced during the survey (284 cfs). However, 1 seine haul was performed, yielding 334 fish. We recorded 220 red shiners, 66 fathead minnows, 1 green sunfish, and no native species.

VERDE RIVER

Horseshoe Reservoir was surveyed with gill nets and by electrofishing on February 25-27, 1998 to document presence of Colorado squawfish and razorback suckers from previous stockings in the Verde River. A total of 561 fish were collected, exclusively nonnatives. A total of 46.5 net night units of effort (1 unit = 1 standard gill net set from sunset to sunrise) were expended, resulting in 456 fish caught and a CPUE of 9.8 fish/net night unit. We expended 114 minutes electrofishing; 105 fish were caught, for a CPUE of 0.92 fish/minute. The 8 nonnative species caught included black crappie, common carp, channel catfish, flathead catfish, bluegill, largemouth bass, smallmouth bass, and goldfish (*Carassius auratus*).

On July 16, 1997, we surveyed the Verde River at the Childs Powerplant campground to evaluate movement and survival of stocked razorbacks and squawfish. A canoe mounted electrofisher was used for 7.3 minutes. We sampled 165 fish; 41 razorbacks and 1 squawfish were collected. Most of these fish were holding in pools and at the base of riffles. However, some were found at the top of riffles, and in slack water at the back of pools. No previously PIT-tagged fish were detected. Twenty-eight (28) of the 41 razorback suckers sampled and the single squawfish were stocked on July 11, 1997. Thirteen razorbacks and the squawfish were PIT tagged (Table 2). Species composition of this survey consisted

of common carp (55.6%), razorback sucker (24.8%), flathead catfish (6.7%), channel catfish (4.2%), Sonora sucker (3.0%), smallmouth bass (3.0%), largemouth bass (2.4%), and Colorado squawfish (0.6%). The CPUE was 2.24 fish/minute. A 15 pound flathead catfish was caught with a razorback sucker (245 mm SL) in its stomach.

Table 2. Lengths, weights, coded wire tag locations, hatchery origin, and PIT tag numbers of Colorado squawfish and razorback suckers collected from the Verde River at Childs on July 16, 1997.

¹ Species	Total length (mm)	Weight (g)	² Wire Tag Location	Origin ³	PIT Tag [*] Number
XYTE	426	720	RD	BPSH	019803858
XYTE	430	900	RD	BPSH	018357011
XYTE	430	1020	RD	BPSH	014820256
XYTE	408	800	RD	BPSH	-
XYTE	463	920	RC	BPSH	015867090
XYTE	476	820	RC	BPSH	015514808
XYTE	495	1060	RC	BPSH	014782873
XYTE	455	960	RC	BPSH	013126600
XYTE	479	1020	RC	BPSH	013000807
XYTE	405	660	RC	BPSH	012035079
XYTE	360	500	No tag	?	014324067
XYTE	523	1080	No tag	?	018885615
XYTE	445	800	No tag	?	012586617
PTLU	496	920	No tag	BPSH	012293088

¹ PTLU=*Ptychocheilus lucius*; XYTE=*Xyrauchen texanus*

² RC=Right Cheek (11/96), RD=Right side of fish under dorsal fin (7/97).

³ All Coded Wire Tagged Fish were stocked in December, 1996.

^{*} All XYTE inoculated with PIT tags at the time of sample.

On August 25 and August 26, 1997, we sampled 3 sites on the lower Verde River between Horseshoe Dam and Bartlett Lake. Two were sampled with a canoe-mounted electrofisher; 1 was sampled overnight with gill nets. We collected 152 fish, consisting of common carp (40.1%), flathead catfish (17.1%), channel catfish (14.5%), threadfin shad (*Dorosoma petenense* [9.9 %]), largemouth bass (8.6%),

green sunfish (7.2%), smallmouth bass (2.0%), and Sonora sucker (0.7%). We sampled 143 fish by electrofisher, expending 93 minutes effort, for a CPUE of 1.5 fish\minute. Gill nets were set at the Devil's Hole site (Bartlett Lake). We collected 10 fish; all were nonnative species (common carp, channel catfish, and flathead catfish). We expended 2.0 net night units of gill net effort, resulting in an overall CPUE of 5.0 fish per net unit.

Another day-long sampling trip at Childs was undertaken on January 7, 1998. Flows were 300 cfs, water was turbid, and pH 8.4. Conductivity was 420 μ S, and temperature was 8°C. Eighty-four (84) fish were caught with 90.6 minutes electrofishing expended for a CPUE of 0.93 fish\minute. Native fishes included 18 razorbacks, 18 squawfish (Tables 3 and 4), and 1 Sonora sucker. Relative abundance was comprised of common carp (46.7%), squawfish (21.4%), razorback suckers (21.4%), largemouth bass (3.6%), channel catfish (2.4%), green sunfish (1.2%), and Sonora sucker (1.2%). Three stocking classes of razorback suckers were represented in this survey. Both species were found occupying pools at the base of riffles, but squawfish also favored runs containing boulders with depths of 1.5 meters. Razorbacks were also found occupying the back of pools immediately above riffles, which were generally shallow (1 m). Seven tuberculate male razorbacks were observed, 2 of which expressed gametes.

Table 3. Lengths, weights, coded wire tag locations, hatchery origin, and PIT tag numbers of Colorado squawfish collected from the Verde River at Childs, January 7, 1998.

¹ Species	Total length (mm)	Weight (gm)	² Wire Tag Location	Origin	PIT Tag [*] Number
PTLU	443	1040	RD	BPSH	114562146A
PTLU	443	650	RD	BPSH	11455727A
PTLU	454	800	RD	BPSH	114527146A
PTLU	487	880	RD	BPSH	114523473A
PTLU	415	510	RD	BPSH	114916743A
PTLU	436	730	RD	BPSH	114526460A
PTLU	474	940	RD	BPSH	11652545A
PTLU	445	720	RD	BPSH	114526790A
PTLU	440	700	RD	BPSH	11454591A
PTLU	412	580	RD	BPSH	114931111A
PTLU	414	610	RD	BPSH	114919385A
PTLU	393	330	RD	BPSH	115624212A
PTLU	397	540	RD	BPSH	114927753A
PTLU	390	490	RD	BPSH	114926510A
PTLU	473	830	RD	BPSH	114915560A
PTLU	453	740	RD	BPSH	114924462A
PTLU	442	830	RD	BPSH	114645096A
PTLU	372	430	RD	BPSH	115613117A

¹PTLU=*Ptychocheilus lucius*; XYTE=*Xyrauchen texanus*

²RD=Right side of fish under dorsal fin (11/96).

^{*}All PTLU and XYTE inoculated with PIT tags at the time of sample.

PTLU = 18 -> 14 minutes at large

Table 4. Lengths, weights, coded wire tag locations, hatchery origin, and PIT tag numbers of razorback suckers collected from the Verde River at Childs, January 7, 1998.

¹ Species	Total length (mm)	Weight (gm)	² Wire Tag Location	Origin	PIT Tag Number
XYTE	411	660	no tag	BPSH	115132777A
XYTE	443	790	LD	BPSH	114932761A
XYTE	447	850	RD	BPSH	114762511A
XYTE	438	800	RD	BPSH	114562747A
XYTE	465	910	no tag	BPSH	115279531A
XYTE	407	740	LD	BPSH	114637227A
XYTE	421	750	RD	BPSH	114554126A
XYTE	395	690	LD	BPSH	114919692A
XYTE	435	950	LD	BPSH	115279251A
XYTE	445	960	no tag	BPSH	114928743A
XYTE	417	750	no tag	BPSH	114669450A
XYTE	420	770	LD	BPSH	115623331A
XYTE	391	620	LD	BPSH	114562447A
XYTE	397	700	LD	BPSH	114652322A
XYTE	355	440	LD	BPSH	114556553A
XYTE	406	670	RD	BPSH	115272590A
XYTE	391	670	N - 35mm	BPSH	114665471A
XYTE	450	920	RD	BPSH	115219544A

¹PTLU=*Ptychocheilus lucius*; XYTE=*Xyrauchen texanus*

² N=Nose (2/95), RD=Right side of fish under dorsal fin (11/96), LD=Left side of fish under dorsal fin (12/97).

* All PTLU and XYTE inoculated with PIT tags at the time of sample.

From January 19 through the 22nd, the Verde River was sampled from Childs to Sheep Bridge. This stretch of the Verde has rarely been sampled during the winter months. The goal of this survey was to detect movement, survival, and perhaps spawning behavior in razorback suckers and Colorado squawfish. Flows were moderate to high at 394 cfs at the gage at Tangle Creek. The water was highly turbid and temperatures ranged from 10-12 °C. The conductivity was 540 µS and

the pH was 8.4. The survey was performed with a canoe mounted electrofisher, 2 support rafts and 1 canoe. A trammel net (45.7 m x 1.8 m x 3.8 cm) was set on 2 separate night camp spots. The river was very wide, and movements of fish were very difficult to detect. A total of 41 fishes were sampled and comprised of Sonora sucker (29.3%), desert sucker (7.3%), smallmouth bass (19.5%), largemouth bass (24.4%), green sunfish (7.3%), common carp (9.8%), and channel catfish (2.4%). A total effort of 226.6 minutes of electrofishing was performed to catch 27 fish. A total effort of 3.3 units of Trammel net effort (1 unit = 100 m² of net for 12 hours) was performed yielding 14 fish. No razorback suckers or Colorado squawfish were sampled.

On May 12-14, 1998, the Verde River from Beasley Flat (RM. 59.5) to Childs (RM.43.8) was surveyed by canoe electrofisher, trammel and seine nets. Flows were 190 cfs at Chasm Creek. Water temperatures were between 16 and 20 °C, conductivity was 660 µS, and pH was 8.9. A total of 134.2 minutes of electrofishing effort was expended to sample 181 fish. CPUE was 1.3 fish\minute. Species sampled by electrofisher included smallmouth bass (7.7%), largemouth bass (7.2%), common carp (32%), channel catfish (6.6%), flathead catfish (7.7%), green sunfish (3.3%), walleye (*Stizostedion vitreum* [0.6%]), Sonora sucker (26%), desert sucker (6.6%), roundtail chub (*Gila robusta*) (1.1%), and Colorado squawfish (1.1%). Three trammel nets were used on 2 overnight sets (2 @ 15.2 m x 1.2 m x 3.8 cm; 1 @ 45.7 m x 1.8 m x 3.8 cm). A total of 1.19 units of trammel net effort was performed to sample 6 fish, for a CPUE of 5.04 fish\TNE. Flathead catfish, channel catfish, and Sonora sucker were the only fish species sampled by Trammel net. A total area of 930.1 m² was sampled by seine net (4.6 m x 1.8 m x .32 cm) yielding 1025 fish and a CPUE of 1.1 fish\m². Red shiner and mosquitofish (*Gambusia affinis*), were sampled by seine net. Two (2) Colorado squawfish were sampled approximately 1 mile downstream of Beasley Flat. They were taken in swift run habitat along an undercut bank. The habitat had sandy, cobble bottom and was about 1.5 m deep. Both squawfish had a coded wire tag in the right side of the fish under the dorsal fin. One squawfish was inoculated with PIT tag number A00081071, measured 450 mm and weighed 690 g. The other squawfish was tagged with PIT tag number A00081211, measured 414 mm and weighed 450 g. Both fish were stocked at Beasley Flat in October, 1997.

Table 5. Lengths, weights, coded wire tag locations, hatchery origin, and PIT tag numbers of razorback suckers and Colorado squawfish collected from the Verde River from Childs to Sheep Bridge, May 26-30, 1998.

¹ Species	Total length (mm)	Weight (gm)	² Wire Tag Location	Origin	PIT Tag Number
XYTE	438	850	RD	BPSH	014820256
XYTE	356	510	LD	BPSH	A00087540
PTLU	440	620	RD	BPSH	A00087367
PTLU	425	570	RD	BPSH	A00083010
PTLU	427	500	RD	BPSH	A00083902
PTLU	497	940	RD	BPSH	A00082540
PTLU	419	490	RD	BPSH	A00081249
PTLU	470	670	RD	BPSH	A00082724
PTLU	411	470	RD	BPSH	A00085038
PTLU	430	550	RD	BPSH	NOT TAG
PTLU	471	620	RD	BPSH	mortality
PTLU	426	600	RD	BPSH	A00083318

¹PTLU=*Ptychocheilus lucius*; XYTE=*Xyrauchen texanus*

²RD=Right side of fish under dorsal fin (11/96), LD=Left side of fish under dorsal fin(12/97).

* All PTLU and XYTE inoculated with PIT tags at the time of sample.

On May 26-30, 1998, the Verde River was sampled from Childs (RM.49.5) to Sheep Bridge (RM. 10) by canoe mounted electrofisher, trammel and seine nets. Flows were 185 cfs at Tangle Creek. Water temperatures ranged from 22 to 25°C, conductivity was 660 μ S, and the pH was 8.3. Ten (10) Colorado squawfish and 2 razorback suckers were sampled (Table 5). A total effort of 290.9 minutes of electrofishing was expended to sample 549 fish which yielded a CPUE of 1.9 fish/minute. Two (2) Colorado squawfish and 2 razorback suckers were sampled by the electrofisher. Other species sampled by electrofisher were common carp (55.4%), smallmouth bass (13.5%), flathead catfish (9.8%), channel catfish (7.7%); green sunfish (5.1%), largemouth bass (2.6%), Sonora sucker (2%), desert sucker (2%), yellow bullhead (0.7%), Colorado squawfish (0.4%), razorback sucker (0.4%), bluegill (0.2%), roundtail chub (0.2%). Sixty-six (66) fish were sampled in 3.97 units of trammel net effort (1 unit = 100 m² of net for 12 hours) by 3 nets (2 @ 15.2 m x 1.8 m x 3.8 cm; 1 @ 45.7 m x 1.8 m x 3.8 cm). CPUE was 16.62 fish/TNE. Eight (8) Colorado squawfish were collected by trammel net, all except

1 were collected at RM 38. There was 1 squawfish mortality. Species sampled by trammel net included common carp (42.4%), flathead catfish (20%), channel catfish (12%), Colorado squawfish (12%), largemouth bass (4.5%), smallmouth bass (3%), green sunfish (3%), Sonora sucker (1.5%), desert sucker (1.5%). A total area of 4373 m² was sampled by seine net (4.6 m x 1.8 m x .32 cm) and yielded 1427 fish for a CPUE of 0.326 fish/m². Species sampled included red shiner, native sucker fry, green sunfish, largemouth bass, mosquitofish, and smallmouth bass. The sucker fry were too small to identify to species.

3. Thirty one (31) Colorado squawfish were collected during 6 monitoring efforts in the Verde River. Again in 1998, no squawfish were sampled in Horseshoe or Roosevelt Lake. Horseshoe Lake was 12 percent full with the lake level rising 25 cm a day during the survey. The lake was highly turbid with substrates consisting of silt, cobble, and boulder. A similar amount of effort was put forth on this survey to that of previous years. Roosevelt Lake was 25 percent full with 4200 cfs entering from the Salt River. Substrates consisted of silt, sand, cobble and boulder. Various depths were sampled in various habitats. Data suggests that Colorado squawfish are moving through the Verde system more evenly than in 1996 when a large group moved to Horseshoe Lake in less than a month. Reports from anglers indicates they have seen and in some cases caught squawfish throughout the Verde from Beasley Flat to Sheep Bridge. Several squawfish have been taken from anglers at Beasley Flat and at Childs by AGFD and local law enforcement. Information signs were placed at Beasley Flat and Childs to educate anglers.

Of the 31 squawfish sampled on the Verde River, 10 were sampled in slow moving pool areas over sand/silt substrates with sedge and/or cattail (*Typha sp.*) lined banks, in less than 2 meters of water. Eight (8) of these were sampled in overnight net sets. Two were collected from backwaters during day sampling. The remaining 21 squawfish, were taken from swift water situations in riffles with cobble/boulder substrates, in less than 1.5 m of water, and at the base of riffle/run areas with depths from 1-3 meters. Sampling data seems to indicate no seasonal variation in squawfish habitat preference. In angler interviews, most squawfish were caught in swift run areas below riffles or rapids, with in-line spinners and crank baits. Some anglers had success catching squawfish near the stocking areas using live night-crawlers.

Sixty-one (61) razorback suckers were sampled from the Verde River between Beasley Flat and Sheep Bridge. Of those razorbacks sampled, 59 were sampled around the Childs area in 1 summer survey and 1 winter survey. All razorbacks sampled at Childs were taken in pools just above or below riffles. Substrates were silt/sand or small cobble with depths ranging from 0.5-4 meters. This did not fluctuate with season or water turbidity. Two (2) razorbacks were sampled below

Childs within 3 miles. These fish were sampled in a run habitat in 2 meters of water over cobble/boulder substrates. They were sampled with 2 squawfish in a narrow run (6 m) with a deep undercut grassy bank. One of these razorbacks had been previously tagged (PIT tag # 014820256). It was first sampled at Childs (RM 43.8) on July 16, 1997. It weighed 1020 g and measured 430 mm at first sample. When recaptured 3 miles downstream (RM. 40.7) it weighed 850 g and measured 438 mm. It still had a coded wire tag under the dorsal fin on the right side of the fish and indicated stocking at Childs in December 1996. This fish was healthy and free of parasites.

In general, parasite loads were very low during this survey year. All except 2 of the 92 razorbacks and squawfish sampled had fewer than 10 active *Lernaea* sp., a parasitic copepod, on their bodies. Two razorbacks had between 20 and 30 *Lernaea* sp. on their bodies. High river flows (>1000 cfs) for extended periods during the spring months, may have reduced parasite loads in the Verde system.

To date, 316 flathead catfish >300 mm have been tagged with Floy type tags in the Upper Salt River. Thirty-one (31) of these have been recaptured, with information on movements and growth patterns recorded. Five (5) of the recaptures have come from anglers who have called and reported their catch. Efforts continue to learn more about the predator load in the Upper Salt River, in hopes of assisting future stocks of native fishes in their survival.

4. No razorback suckers or Colorado squawfish were collected for museum or laboratory analysis. One squawfish mortality was examined after rejuvenation attempts had failed. Stomach contents were analyzed, and documented.
5. All sensitive species site-specific locality and status data was submitted to the Heritage Data Management System for use in Section 7 clearances and other consultation with the U.S. Fish and Wildlife Service.
7. We prepared this performance report summarizing all activity under this Job.

B. Recommendations:

Salt River:

Continued efforts should be made to reach agreement with the White Mountain Apache Tribe to resume razorback sucker and Colorado squawfish stocking into the Salt River. The objective should remain 2000 fish (>300mm) of each species stocked at Gleason Flat by DNFHTC, until a need for change is identified. Methods and feasibility of control or suppression of nonnative species, particularly large predatory catfish, should be continued in portions of the upper Salt River.

A management plan should be developed for the Salt River flathead catfish fishery. Native species are now largely extirpated from this river. Reintroduction of Sonora sucker, desert sucker, and roundtail chub, combined with balanced catfish management, might help reintroduction efforts for razorback suckers and Colorado squawfish by increasing species diversity. Possible control of the red shiner should also be investigated.

Site-specific surveys should be planned with access and repeatability in mind. A minimum of 3000 seconds per day electrofishing, 10 seine hauls, and 3 net sets or set fishing lines per day should be performed. Until stockings resume at Gleason Flat and/or the U.S. 60 highway bridge, surveys should be performed once per year above Horseshoe Bend.

Verde River:

Minimum efforts, stratified within reaches, should be established for monitoring efforts on the Verde River. Future Verde River monitoring efforts should include a minimum of 10 seine hauls per day, a minimum of 2 gill/trammel nets set each night, and at least 3000 seconds of electrofishing per day. Stocking of 2000 large razorback suckers (>300 mm) per year into the Verde River at Childs by BPSH should be continued. A minimum of 2000 large (>300 mm) Colorado squawfish should continue to be stocked into the Verde River.

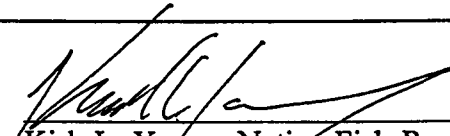
Efforts should be initiated to document/quantify parasite loads in the Verde system, and further determine potential impacts on reintroduced razorback suckers and Colorado squawfish.

C. Significant Deviations:

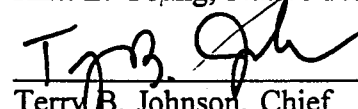
DNFHTC did not stock Colorado squawfish and razorback suckers into the Salt River at Gleason Flat or the U.S. 60 highway bridge, due to concerns expressed by the White Mountain Apache Tribe.

Due to lack of stocking, logistical constraints, and the need to focus activities in areas of stocking, we did not survey the Verde River from Perkinsville to Clarkdale. We surveyed twice at Childs to detect dispersal of stocked razorbacks and squawfish, and once in winter survey from Childs to Sheep Bridge to detect possible spawning activities of razorback suckers.

D. Prepared by:


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